Solutions by **Andrew Lys** `alys(at)ncsu.edu`

Problem 1. Consider

$$y = \sigma(w^\top x)$$
$$\ell = \mathcal{L}(y)$$

for $w, x \in \mathbb{R}^d$ and $\sigma(z) = z^+$. Gve an expression for $\frac{\partial \ell}{\partial w_i}$ as a function of $\frac{d\mathcal{L}(y)}{dy}$.

Proof. By the chain rule, we have:

$$\begin{aligned}\frac{\partial \ell}{\partial w_i} &= \frac{d\mathcal{L}(y)}{dy}\frac{\partial y}{\partial w_i} \\ &= \frac{d\mathcal{L}(y)}{dy}\frac{\partial \sigma(w^\top x)}{\partial w_i} \\ &= \frac{d\mathcal{L}(y)}{dy}\sigma'(w^\top x)x_i \\ &= \frac{d\mathcal{L}(y)}{dy}\mathbb{1}_{w^\top x > 0}x_i\end{aligned}$$

□

Problem 2.

$$\begin{aligned}&\text{for } b,j \quad s.\text{grad}[b,j] += p.\text{grad}[b,j]\exp(s[b,j])/Z[b] \\ &\text{for } b,j \quad Z.\text{grad}[b] -= p.\text{grad}[b,j]\exp(s[b,j])/(Z[b])^2 \\ &\text{for } b,j \quad s.\text{grad}[b,j] += Z.\text{grad}[b]\exp(s[b,j])\end{aligned}$$

Problem 3. Consider the following code:

$$\begin{aligned}&\text{for } b,j \quad e[b] -= p[b,j]p.\text{grad}[b,j] \\ &\text{for } b,j \quad s.\text{grad}[b,j] += p[b,j](p.\text{grad}[b,j] + e[b])\end{aligned}$$

Show how this code computes the same gradient as the code in Problem 2.

Proof. We can plug in $Z.\text{grad}[b]$ from the second line of Problem 2 into the third line of Problem 2 to get:

$$\begin{aligned}\text{for } b,j \quad s.\text{grad}[b,j] += &\left(\sum_j -p.\text{grad}[b,j]\exp(s[b,j])/(Z[b])^2\right)\exp(s[b,j]) \\ = &\frac{e[b]}{Z[b]}\exp(s[b,j]) = p[b,j]e[b]\end{aligned}$$

Additionally, we may rewrite the first line of Problem 2 as:

$$\text{for } b,j \quad s.\text{grad}[b,j] += p.\text{grad}[b,j]\exp(s[b,j])/Z[b] = p[b,j]p.\text{grad}[b,j]$$

Now that we have no reliance on $Z.\text{grad}[b]$, we can combine the two lines, after computing $e[b]$, to get:

$$\text{for } b,j \quad s.\text{grad}[b,j] += p[b,j]p.\text{grad}[b,j] + p[b,j]e[b] = p[b,j](p.\text{grad}[b,j] + e[b])$$

□

Problem 4.

Part (a).

$$\begin{aligned}\text{for } b,j,k \quad &\tilde{R}_t[b,j] += W^{h,R}[j,k]h_{t-1}[b,k] \\ \text{for } b,j,k \quad &\tilde{R}_t[b,j] += W^{x,R}[j,k]x_t[b,k] \\ \text{for } b,j \quad &\tilde{R}_t[b,j] -= B^R[j]\end{aligned}$$

Part (b).

$$\begin{aligned}
\text{for } b, j \quad & B^R.\text{grad}[j] -= \frac{1}{B}\tilde{R}_t.\text{grad}[b, j] \\
\text{for } b, j, k \quad & W^{x,R}.\text{grad}[j, k] += \frac{1}{B}\tilde{R}_t.\text{grad}[b, j] x_t[b, k] \\
\text{for } b, j, k \quad & W^{h,R}.\text{grad}[j, k] += \frac{1}{B}\tilde{R}_t.\text{grad}[b, j] h_{t-1}[b, k] \\
\text{for } b, j, k \quad & h_{t-1}.\text{grad}[b, k] += \tilde{R}_t.\text{grad}[b, j] W^{h,R}[j, k]
\end{aligned}$$

Problem 5.

```
hs = []
hs.append(CELL(Phi, X[0], ZERO))
for t in range(1, T + 1):
    hs.append(CELL(Phi, X[t], hs[-1]))
```